

EXP NO – 18

Gas Sensor MQ-2 in Proteus using Arduino

AIM

To design and simulate a gas leakage detection system using an MQ-2 gas sensor and Arduino UNO in Proteus. When gas is detected, an LED and buzzer are activated.

PROGRAM

```
int gasSensor = 2;
int led = 8;
int buzzer = 9;

void setup()
{
    pinMode(gasSensor, INPUT);
    pinMode(led, OUTPUT);
    pinMode(buzzer, OUTPUT);

    digitalWrite(led, LOW);
    digitalWrite(buzzer, LOW);

    Serial.begin(9600);
}

void loop()
{
    int gasState = digitalRead(gasSensor);

    if (gasState == LOW)
    {
        digitalWrite(led, HIGH);
        digitalWrite(buzzer, HIGH);

        Serial.println("GAS DETECTED!");
    }
    else
    {

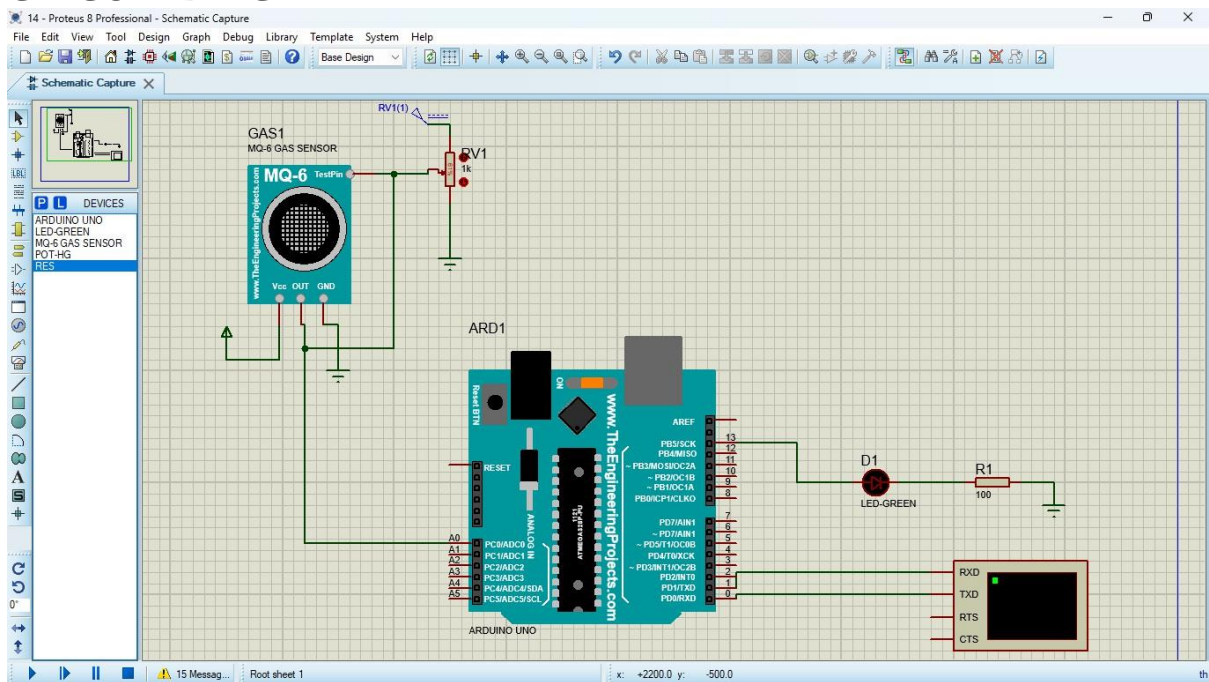
```

```
digitalWrite(led, LOW);
digitalWrite(buzzer, LOW);
```

```
Serial.println("NO GAS");
}
```

```
delay(500);
}
```

CIRCUIT DIAGRAM



OUTPUT

